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State-of-the-art review on power grid resilience to extreme weather events: Definitions, frameworks, quantitative assessment methodologies, and enhancement strategies

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HIGHLIGHTS

- Different definitions of power grid resilience are studied and clarified.
- Relation of extreme event and extreme impact to power grid resilience are discussed.
- Power grid resilience framework is explained.
- Existing power grid resilience assessment methodologies and indices are described.
- Power grid resilience enhancement strategies are presented.

ARTICLE INFO

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Grid resilience definition
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Grid resilience index
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ABSTRACT

The rise of power outages caused by extreme weather events and the frequency of extreme weather events has motivated the study of grid resilience. This paper presents a state-of-the-art review of existing research on the study of grid resilience, which focuses on the point of view of power system engineering with respect to extreme weather events. Firstly, it investigates confounding terminologies used in the study of grid resilience, such as the definitions, the differences with grid reliability, the extreme weather events, and their extreme impact on the power systems. Secondly, it presents a grid resilience framework as a general provision to understand the subjects in the study of grid resilience. Thirdly, it describes several methodologies of grid resilience assessment and some quantitative indices. Finally, various grid resilience enhancement strategies implementations are discussed.

1. Introduction

The frequency and severity of power outages caused by extreme weather events have increased, which have led to the enormous damages to people and the economy. In the United States, seven out of the ten years between 2003 and 2012 had more than 10 million customers affected by weather-related outages [1]. A 2012 survey by the U.S. Energy Information Administration revealed that the observed outages to the bulk electric system in the United States from 1992 to 2012 show a significant increase of the outages caused by weather-related events [2]. The data for large blackouts caused by extreme weather events and their trends in the United States is disclosed by [3,4]. A notable increase of grid disturbances caused by the weather was also observed in the Nordic region, as reported by the European Network of Transmission System Operators for Electricity (ENTSO-E) [5]. The report summarized the grid disturbances in five countries, including Denmark, Finland, Iceland, Norway, and Sweden, and showed that 30–60% of the outages resulted from environmental factors, which were the most significant outages causes. Some of the major weather-related power outages worldwide from 2011 to 2016 are presented in Table 1. At the same time, climate change has caused an increase of extreme weather events occurrences [6] and thus, it will increase the likelihood of the extreme impacts on the power systems. These facts motivate many countries to study the resilience of the electrical power grid, which will be called as "grid resilience" hereafter. For example, the United States have placed an added emphasis on R&D to enhance the resilience to climate change and extreme weather [7], and the U.K. National Security Strategy has set out a guide to improve the resilience of critical and essential services

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Table 1
Major power outages worldwide (2011–2016).

Date	Location	Extreme weather event	Number of customers without power
September 2016	South Australia, Australia	The Blyth Tornado	1,700,000
September 2016	Florida, United States	Hurricane Hermine	300,000
November 2015	Washington, United States	Windstorm	> 161,000
November 2015	Vancouver, Canada	Windstorm	700,000
July 2014	Luzon, Philippine	Typhoon Rammasun	13,000,000
December 2013	Ontario, Canada	Ice Storm	~300,000
April 2013	Poland	Heavy snow	100,000
March 2013	Greater Belfast, Northern Ireland	Heavy snow	200,000
February 2013	Northeastern Area, United States	The North American Blizzard	650,000
January 2013	SE Queensland, Australia	Cyclone Oswald	250,000
October 2012	Eastern Area, United States	Hurricane Sandy	8,100,000
July 2012	Northern Area, India	Monsoon	> 600,000,000
June 2012	Ohio, United States	Thunderstorm	3,800,000
February 2011	Christchurch, New Zealand	Earthquake	~160,000

against natural hazards [8].

The scopes of grid resilience are large and cover many aspects, such as technical, economic, social, and policy. Technically, it involves multi-disciplinary knowledge, such as power system, civil and structure, geography, computer science, probability, and meteorology and climatology. The study of grid resilience also relates to various organizations such as utility companies, disaster management agencies, and national weather services. The interdependency between the grid and other sectors, such as water, fuel, and transportation, must also be considered in the study of grid resilience. However, the influence of interdependency with other areas is beyond the scope of this review. Although the study of grid resilience has gained popularity in the recent years, it is still not well established, and research and development in this area is still being carried out by many institutions.

The motivation of this paper lies on the need to present a comprehensive study of grid resilience, which focuses on the technical matters from the point of view of power system engineering with respect to extreme weather events. This paper gives an overview of confounding terminologies in the study of grid resilience, which cover the not-yet-standardized definitions of grid resilience, the difference between grid resilience and grid reliability, and the interpretation of extreme weather events and their extreme impact on power systems. A grid resilience framework, which provides a general understanding of the study of grid resilience, is also presented. Subsequently, this paper discusses the assessment methodology and some quantitative indices of grid resilience. These indices can be used to evaluate the resilience of the grid, to compare one grid with the others, and to decide what actions should be taken. Lastly, various enhancement strategies to improve the grid resilience are discussed. Non-technical aspects which are also equally important in the study of grid resilience, such as economic, social, and policy, are not reviewed in this paper and will be considered in future works.

The rest of this paper is organized as follows: Section 2 describes the concept of grid resilience. Section 3 presents the framework of grid resilience. Section 4 explains the assessment methodology in grid resilience and some quantitative indices. Section 5 discusses the grid resilience enhancement strategies, and the conclusion and future works are summarized in Section 6.

2. The concept of grid resilience

2.1. The definition of grid resilience

The concept of resilience has been studied in several fields such as social, community, and ecological systems [9–12], risk management and disaster adaptation [13–15], economic systems [16], transportation systems [17], and organization management [18]. The word resilience comes from the Latin word, *resilio*, which literally means to 'spring back' [14]. The contextual meaning of resilience is the ability of a system to recover quickly from disruption. The terminology of grid resilience or power system resilience started to develop along with the study of the resilience of critical infrastructures. Therefore, it is necessary to define the general meaning of resilience to have a better understanding of grid resilience.

In their 2012 report, the Intergovernmental Panel on Climate Change (IPCC) [19] described resilience as a system's ability to anticipate, absorb, and recover from the effects of hazardous events in a timely and efficient manner. A similar definition was also presented by the U.K. Cabinet Office [8], which stated that resilience is a system's ability to anticipate, absorb, adapt to, and rapidly recover from the disruptive event. The U.S. Presidential Policy Directive [20] also remarked on the ability to withstand and recover rapidly from the disruptions. It also added that a system requires the preparation and adaptation capability to face the changing conditions. The U.K. Energy Research Centre (UKERC) [21] construed the definition of energy system resilience as the capacity to tolerate disturbances including the ability to speedily recover from shocks and continue to deliver affordable energy services to customers. The United Nation – International Strategy for Disaster Reduction (UN-ISDR) [22] defined resilience as a degree of a system's adaptability to maintain its functionalities to cope with hazards by organizing and learning from the past disasters.

Some typical properties can be extracted by further examination of the definitions given in the literature. In [8], the component of infrastructure resilience included resistance, reliability, redundancy, and response and recovery. Resistance focuses on providing protection, reliability ensures the system can operate within abnormal conditions, redundancy is the backup of the system, and response and recovery aim at bringing the system back to its normal state. Willis and Loa [23] highlighted four aspects of the system resilience such as the state of the service in response to disruptions, design, and operation, responses of the system, and timescale. Woods [24] presented the concepts of resilience as rebound, robustness, elastic extensibility, and sustained adaptability. The Multidisciplinary Center for Earthquake Engineering Research (MCEER) [25] characterized the dimensions of resilience into robustness, redundancy, resourcefulness, and rapidity, which are well-known as the "4R" dimension of resilience. Lastly, Madni and Jackson [26] described resilience as a multi-faceted capability, including avoiding, absorbing, adapting to, and recovering from disruptions.

The summary of the generic definitions of resilience and its key properties is presented in Table 2.

The definitions mentioned above, which are intended for other systems or infrastructures resilience, can be consulted to understand the meaning of grid resilience [27]. Hence, a resilient grid can be described as a grid which has four fundamental properties of resilience, namely anticipation, absorption, recovery, and adaptability after the damaging events. Anticipation is the ability to avoid any possible damage caused

Table 2
Summary of resilience definitions and its key properties.

Organization	Definition of resilience	Properties of resilience	Ref.
Intergovernmental Panel on Climate Change (IPCC)	"The ability of a system and its component parts to anticipate, absorb, accommodate, or recover from the effects of a hazardous event in a timely and efficient manner, including by ensuring the preservation, restoration, or improvement of its essential basic structures and functions."	Anticipation, absorption, accommodation, timely and efficient recovery	[19]
U.K. Cabinet Office	"The ability of assets, networks, and systems to anticipate, absorb, adapt to and/or rapidly recover from a disruptive event."	Anticipation, absorption, adaptation, rapid recovery	[8]
Presidential Policy Directive/PPD-21	"The ability to prepare for and adapt to changing conditions and withstand and recover rapidly from disruptions. Resilience includes the ability to withstand and recover from deliberate attacks, accidents, or naturally occurring threats or incidents."	Preparation, adaptation, withstanding, rapid recovery	[20]
U.K. Energy Research Centre (UKERC)	"The capacity of an energy system to tolerate disturbance and to continue to deliver affordable energy services to consumers. A resilient energy system can speedily recover from shocks and can provide alternative means of satisfying energy service needs in the event of changed external circumstances."	Tolerance, speedily recovery, provision of alternative service	[21]
United Nation – International Strategy for Disaster Reduction (UN-ISDR)	"The capacity of a system, community or society potentially exposed to hazards to adapt, by resisting or changing to reach and maintain an acceptable level of functioning and structure. This is determined by the extent to which the social system is capable of organizing itself to increase this capacity for learning from past disasters, for better future protection and to improve risk reduction measures."	Resistance, maintain the functionality, learning from the past	[22]
Multidisciplinary Center for Earthquake Engineering Research (MCEER)	"The ability of social units (e.g., organizations, communities) to mitigate hazards, contain the effects of disasters when they occur, and carry out recovery activities in ways that minimize social disruption and mitigate the effects of future disasters."	Mitigation, recovery	[25]

by an extreme weather event. Absorption is the ability of the grid to minimize the damages caused by an extreme weather event. Recovery is the ability of the grid to rebuild its damaged functionalities caused by an extreme weather event. Adaptability is a process whereby a system learns from the past events, to improve its capability, and to be ready to deal with the next event.

2.2. Comparison between grid resilience and grid reliability

The concept of grid resilience relates with the existing notion of grid reliability in power system. A grid will not be considered as resilient if it is not reliable, and indeed, a resilient grid will improve the grid reliability [28]. Grid reliability is a commonly used methodology to evaluate the performance of a power system. It is defined as an grid's ability to perform its function under given conditions and for a given time interval [29,30]. IEEE described reliability as the amount of momentary and sustained interruptions in the systems for a specific time range. IEEE [31] characterized grid reliability in several indices such as System Average Interruption Frequency Index (SAIFI), System Average Interruption Duration Index (SAIDI), and Average Service Availability Index (ASAI). Other references about grid reliability evaluation can be found in [32–34].

In fact, grid reliability study is only examining the frequency and duration of the outages caused by common failures. Common failures are caused by the events with high probability, shorter duration, and smaller affected region, such as line to ground short circuits or device malfunctioning [35,36]. When the outage duration is less than five minutes, it is often excluded from the calculations of grid reliability indices, which means the grid is considered sufficiently reliable [31]. If the duration is longer than five minutes, it is considered in the grid reliability indices calculations. Necessary actions need to be taken to improve grid reliability if the grid is considered less reliable. The improvement actions are carried out to improve the passive operational capability of the grid and focus on the affected area only. However, an extreme weather event may cause a whole system to collapse and the operation, control, protection, and communication capability of the grid may be lost. In this case, the grid requires more active operations to recover its functionalities. Therefore, the impact of an extreme weather event is often excluded in the grid reliability assessment and it is treated separately.

Some authors studied the effects of the weather on the power grid reliability. Ward [37] summarized the effects of the weather on the grid in three weather conditions, namely normal, adverse, and extreme. He also explained that the faults due to normal weather have no significant effect on electricity supply to customers, while if they are caused by adverse or extreme weather, the consequences are severe since it requires physical recovery, which is time-consuming and very costly. Billinton and Wenyuan [38] and Bhuiyan and Allan [39] proposed a method to evaluate the system adequacy by considering the effects of the weather. They divided the geographical area into separate regions, based on the weather conditions instead of assuming a whole geographic area experienced the same weather conditions. Similarly, Wang and Billinton [40] investigated the distribution reliability by considering the weather conditions, and restoration capability in that weather condition. They concluded that the weather not only affects the functionality of the system but also the restoration time, which leads to longer interruption periods on the customer sides. Brown et al. [41] developed a model for storm events and the failure rates during the storms, and proposed a different index for grid reliability called Storm Average Interruption Duration Index (STAIDI). Meanwhile, Liu and Singh [42] evaluated the power system reliability due to a hurricane by considering its spatial and temporal characteristics. Other studies on the evaluation of the effects of hurricanes on grid performances are also discussed in [43–46].

These studies have shown various approach to assess the grid reliability under extreme weather events, but they did not include the effect of the recovery actions into the assessment. Therefore, a new method to evaluate the performance of electricity grids that takes into account extreme weather events and the recovery actions, so-called grid resilience, is required. The assessment of grid resilience is time-dependent since it is evaluated based on the occurrence of extreme weather events. In addition, grid resilience must be able to represent the grid performance in terms of the grid's ability to anticipate, absorb, recover, and adapt to the issues caused by extreme weather events. In grid resilience, rather than operating passively such as in grid reliability, the grid is expected to operate in active ways to cope with the

effects of an extreme weather event. Because of these noticeable differences, although one grid may be evaluated as reliable, it is not necessarily considered resilient. Liu [47] made a distinction between grid reliability and grid resilience based on the type of the events it can withstand. According to this distinction, grid reliability studies to the outages caused by simple, the restoration is straightforward, and the outage durations are short. Whereas, grid resilience concerns with the outages caused by extreme weather events so that the restoration is more complex with a longer outage duration.

Differences in the description of grid resilience and grid reliability are also described in [48], where the authors argue that grid reliability deals with high probability and low impact events, while grid resilience deals with low probability and high impact events. In the study of grid reliability, the evaluation is focused on the system states. While in the study of grid resilience, the evaluation also includes the transition times between states (from normal to abnormal, and vice versa). Furthermore, the concern of grid reliability is the customer interruption time, while grid resilience also considers the infrastructure recovery time.

2.3. Extreme weather event and extreme impact

One of the reasons to study grid resilience is the difficulty in forecasting extreme weather events. The forecast of ordinary weather, which causes frequent failures, has become very accurate. The forecast of this type of weather is considered in every day operation of power system grids. On the contrary, the extreme weather events forecasting is still challenging due to their rare occurrences and lack of historical data.

Extreme weather is defined as the occurrence of a weather variable value above (or below) a threshold value, which is near the upper (or lower) end of the range of observed variable values [19]. Other studies defined it as a rare occurrence of magnitude in the upper or lower part of the variability scale [49]. On the other hand, an extreme impact is defined as a highly significant and long-lasting consequence to the system [19].

The extreme weather event may cause an extreme or non-extreme impact. Likewise, the extreme impact may happen from a common fault that turns into cascading failures and causes a system collapse. In this paper, the study of grid resilience is emphasized on the performance of the grid under extreme weather events, instead of any normal weathers that turn into an extreme impact. Grid resilience deals with the extreme weather events whose likelihood is low but their probability to induce massive and extensive impact is high. Therefore, a large impact caused by non-extreme weather event is not considered in this paper. Fig. 1 shows the scope of grid resilience based on the type of weather. It indicates that grid resilience focuses on the extreme weather events, rather than extreme impacts. The figure maps the impact caused by the weather, which results from three factors including the intensity of the weather, grid exposure, and grid vulnerability [50–52].

2.3.1. Weather intensity: extreme and normal weather events

The extreme weather cannot be controlled owing to its natural behaviour, while extreme impacts can be controlled through management of grid exposure and grid vulnerability. The severity of the extreme weather events is classified based on its intensity (magnitude), duration, and the area it affects. For instance, the storms, hurricanes, and cyclones are mostly evaluated based on their contained-wind speed. Supplementary parameters include the radius of maximum wind, to estimate the highest rainfall rates, and the radius of outermost closed isobar (ROCI). The scale for the contained-wind speed of the storms, hurricanes, and cyclones is outlined in the Beaufort wind scale or the Saffir-Simpson Hurricane Wind Scale [53,54]. Earthquakes on the other hand, are usually measured by Richter scale, supplemented by the location of its epicenter and depth.

Attempts to classify the weather into extreme or non-extreme have been proposed in the literature. Previously, extreme events were qualitatively categorized based on the survey conducted by the Edison Electric Institute (EEI). For instance, an extreme event is defined as any interruptions, resulting from a catastrophic event, that exceeds the design limits of the electric power system. Examples of extreme events are earthquakes, tornados, and extreme storms. In this definition, there was no specific number to evaluate the weather and the amount of disruption. Another work defined a major event as one which causes 10% or more of customers to be without power for more than 24 h. These definitions were not uniform, and the categorization ignored the size of the system, as well as the cause of the disruption. Therefore, it was difficult to apply this to the different power systems.

The IEEE provided another methodology to determine major events [31], whereby the power outages caused by extreme weather events are excluded and treated separately from the reliability study. This event is categorized as Major Event Day (MED). MEDs are defined as days with larger SAIDI values. The classification of MED is based on the pre-determined threshold index obtained using 2.5 beta methodology. Days having SAIDI greater than the threshold value are considered as MEDs.

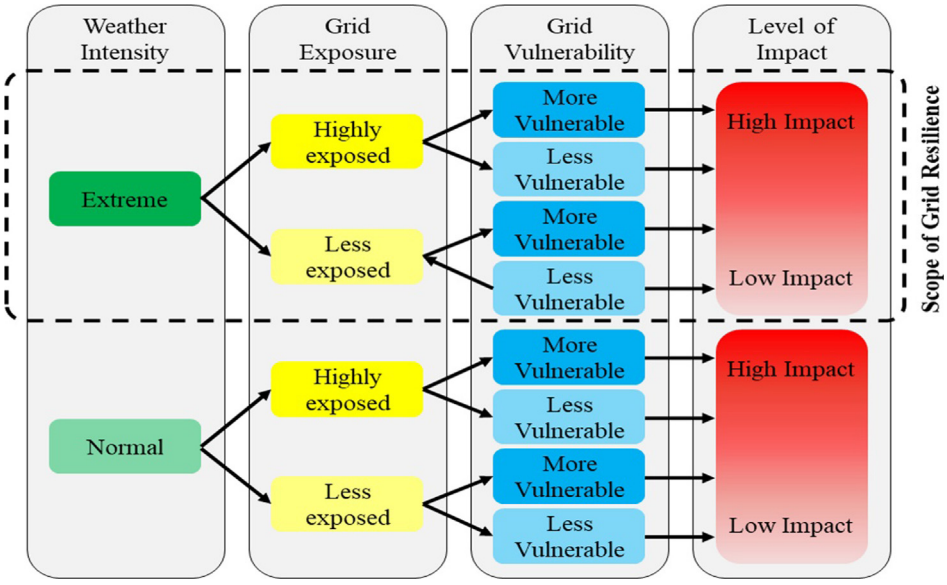


Fig. 1. The scope of grid resilience.

These MEDs can be utilized as reference data to study grid resilience.

An IPCC report [19] suggests another methodology to classify the weather events. It defines 'extreme climate' or 'weather event' or 'climate extreme' as "the occurrence of a value of weather or climate variable above (or below) a threshold value near the upper (or lower) ends of the range of observed values of the variable." This method requires two parameters to categorize the extreme weather events, namely the occurrence and a specific threshold.

2.3.2. Grid exposure: the presence of the grid in the hazard area

The presence of extreme weather event will not be an issue if they occur in a location where grids or parts of the grid are not present, and it will have less impact if only small parts of the grid are located where the extreme weather even occurs. The IPCC [19] defined grid exposure as the presence of people, livelihoods, or systems in a location that can be affected by an extreme weather event. Grid exposure is also defined as the number of people, properties, or systems in hazardous areas, so they subject to potential losses [55]. Crichton [56] described the grid exposure as the existence of properties in a risky area, where there is a possibility of damage. It is also described as the amount of uncovered or unprotected parts (bareness) of the grid present at the location of extreme weather events.

The amount of exposed grid is weather-dependent. The weather will affect a wider area when its intensity is high. Hence, if a wider area is hit by the weather, the amount of exposed grid will be larger. Grid exposure can be quantified by the number of grids or grid parts in a specific geographical area where an extreme weather event occurs. An obvious way to reduce grid exposure is to relocate the grid or grid parts from the hazardous area, but this would be very costly to implement. Therefore, rather than relocating the grid, preparing the grid to be weather-resistant is more feasible.

2.3.3. Grid vulnerability: the evaluation of the weakness of the grid

Grid vulnerability expresses the weaknesses of the grid. In [56], vulnerability is described as the likelihood of a property to get damaged due to an extreme event. Furthermore, it defines the physical and operational characteristics possessed by a system that determine the susceptibility of that system to the impacts of hazards [57]. Birkmann [58] explained the vulnerability as a crucial property in the determination of the impact. Grid vulnerability can be evaluated regardless of the type and intensity of the extreme weather events. It focuses on the grid capability rather than on the extreme weather event.

Some authors in the literature proposed some methodologies to assess grid vulnerability based on topology and grid operation. Daouda et al. [59] evaluated grid robustness based on the criticality of the network structure and the dynamic behavior when part(s) of the network is damaged. Afgan et al. [60] studied the failures of the transmission line by identification of the weak links in the network. Even though these papers presented the importance of the grid vulnerability, other dimensions of resilience, such as the impact and recovery speed upon damage, were not examined. Other authors assessed the relationship between network topology and reliability of the European power grids [61–65], and a similar approach was used by Kim et al. [66] to examine South Korean power grids. They employed graph theory to evaluate the structural grid vulnerability. The critical level of nodes and lines was measured by the degree, clustering coefficient, and betweenness. Mohagheghi and Javanbakht [67] also analyzed the weak links of the grid and considered their reinforcement a priority.

Modified graph theory is used by other literature to examine the structural vulnerability of the grid. Chen et al. [68] studied the grid vulnerability on random and intentional attacks using a hybrid approach of graph theory, whereby power flow and load shedding were employed in the analysis, rather than graph parameters such as betweenness and node efficiency. A weighted network clustering analysis for general networks was conducted by Tore and Panzarasa [69]. Work of Wang et al. [70] considered the generators and loads characteristics in the analysis of network's betweenness for the vulnerability assessment. They found damages like loss of load when parts of the grid were affected. Power Transfer Distribution Factor (PTDF) was utilized to give weight to the nodes and links to distinguish the criticality of the nodes and links for different configurations [71–74]. The parameters used were a modification of the graph theory parameters such as entropic degree and the net ability of the nodes. Rocchetta and Patelli [75] presented a framework to assess the vulnerability of a power grid by quantifying the uncertainty, which is described using normal distribution, where the parameter is estimated from historical records of load demand in each node of the system. The value for the vulnerability was determined by power flow analysis and graph spectral analysis of the system. Based on the results, the ranking of vulnerability of each component can be determined for further assessment. These works presented an approach to analyze the importance of the structures of the grid, which is required for the assessment of grid resilience. However, further studies which include the restoration, adaptability, and dynamic states of the grid are required.

The operational vulnerability was studied by Eshghi et al. [76]. They measured real-time parameters of the grid to calculate the vulnerability of the grid or parts of the grids. A similar concept was also used by Sun et al. [77]. To the same end, Liu [78] used the failure probability of the transmission line due to operational causes, such as overloading and under-voltage, while Zhan et al. [79] used operational sensitivity. The indices are bus voltage, line flow, and system frequency. The grid is evaluated based on its ability to maintain the grid stability when one or more elements are out of operation.

3. Grid resilience framework

A grid resilience framework presents a general provision on the study of grid resilience. In [48], a long-term resilience framework focused on vulnerability/adaptability analysis, identifying and prioritizing resilience enhancement strategies, cost/benefit analysis, and applications of the enhancement strategies. The vulnerability analysis considers three system states, namely resilient, degraded, and restorative. Cost/benefit analysis was included to classify the enhancement strategies based on resilience or cost-efficiency. Therefore, the final decision can be taken based on the decision of the owner.

Francis and Bekera [80] introduced the resilience assessment framework for engineered systems. The framework starts from system identification, which includes the definition of the system domain, description of fundamental and system objectives, characterization of physical, chemical, spatial, or social properties, and identification of analytical goals and objectives. Subsequently, the assessment requires vulnerability analysis which considers the likelihood of disruptive events and the temporal dynamic behavior. Resilience objective setting is then set by describing the normal system function which should be achieved by the study. After that, the coordination between parties involved in the resilience study is referred to as stakeholder engagement. The last component of this framework is resilience capacities, which consist of three pillars: absorptive, adaptive, and restorative capacities.

Arghandeh et al. [81] proposed a resilience assessment framework for power systems by modifying the so-called conceptual Pressure and Release (PAR) risk framework. System identification includes topology, physical characteristic, operational constraints, and dynamic behavior. Vulnerability analysis is defined as a function of disturbing events, consequences/impacts, system adaptability, and recovery speed (before, during, and after disturbance). On the other hand, resilience operation considers recovery potential and absorbing potential. In [82], a framework for establishing critical infrastructures including the electricity grid was presented, which consists of resilience goal development, sector applications, and resilience improvement. They also mentioned four main principles of resilience, namely robustness (absorbability), resourcefulness (real-time crisis management), rapid

recovery, and adaptability (uptake of lessons learned).

The grid resilience frameworks presented in the literature lead to two typical structures: grid assessment and grid enhancement. In the grid assessment process, the weather and the grid information are utilized to evaluate the conditions of the grid. In addition, the grid is assessed through four conditions which represent each property as described in Section 2.1. They are the prevention state (anticipation), degradation state (absorption), restoration state (restoration), and adaptation state (adaptability). The grid resilience assessment includes the hazard modeling and results in grid resilience indices, which can be used to evaluate the grid condition, to compare one grid to another, and to determine necessary actions to enhance the grid resilience. The grid resilience enhancement strategies on the other hand, can be implemented by referring to the grid resilience indices, along with cost-benefit analysis and stakeholder engagement. The enhancement strategies can be implemented on physical and operational systems of the grid. The grid resilience enhancement is aimed at decreasing the level of the damage caused by an extreme weather event as well as at reducing the time required to bring back the normal functionality of the grid, i.e. the restoration time. The framework of grid resilience is shown in Fig. 2.

4. Grid resilience assessment

Grid resilience can be assessed based on either the impact it experiences due to an extreme weather event or the grid capability it possesses. The impact indicates the amount of damage of the grid, while the grid capability indicates the ability of the grid to maintain its functions in the damaged state and to restore the damaged parts as quickly as possible [83]. Other studies have shown that the grid resilience can be assessed by combining the impact analysis and the grid capability. Espinoza et al. [36] proposed a multi-phase assessment method to evaluate the grid resilience such as threat characterization, vulnerability assessment, system reaction, and system restoration. In addition, they presented the multi-phase adaptation which considers system robustness, redundancy, and response.

As previously mentioned, one of the most challenging issues in the assessment of grid resilience to the extreme weather events is the limitation and the scarcity of historical data, which makes the modeling of the impact caused by an extreme weather event complicated. An extreme weather event may occur once or more in a year, or it may not occur in a couple of year. Furthermore, the occurrence of an extreme weather event does not always cause a significant impact to the grid, making the recorded data not so useful for the model development. Some authors attempted to solve this issue through a statistical approach. The Monte Carlo simulation is used in [84–86] to assess the capability of the grid and to determine the amount of impact caused by extreme weather. It presented a case study of grid resilience assessment by analyzing the impact caused by the extreme weather events as well as the enhanced capability of the grid. It used the fragility model to determine individual components failure probability and used it to calculate the impact in term of Loss of Load Frequency (LOLF) and Loss of Load Expectation (LOLE). Furthermore, defensive islanding was implemented to enhance grid resilience, and the reduction of the impact was calculated. Non-Homogeneous Poisson Process (NHPP) is used to model the impact caused by an extreme weather event [87]. It is differentiated from the impact caused by the weather-independent effects, such as aging, malicious attacks, or manufacturing errors, which is modeled through Homogeneous Poisson Process (HPP).

The impact and the grid capability can be assessed through the grid conditions during the occurrence of an extreme weather event. Ouyang et al. [88–90] analyzed the grid conditions under an extreme weather event in three states. These are disaster prevention, which is the state of the grid before the event, damage propagation, which is the state of the grid between the first occurrence of the event up to the worst state of the grid, and recovery, which is the post-event state. In [35,91], the damage propagation was divided into disruption state and disrupted state. The disruption state refers to the condition when the damaging event is still in progress, and the disrupted state refers to the condition when the damaging event has stopped, but the restoration has not started.

Kwasinski [92] proposed metrics and a quantitative model for the distribution of grid resilience. He presented the metrics for the separate properties of resilience including withstanding capability, recovery speed, preparation capacity, and adaptation capability. The adaptation capability was included because grid resilience is a continuous process that requires the grid to prepare for the next upcoming extreme weather event. The grid must be able to learn from the previous events and apply necessary actions to improve its resilience.

4.1. Grid conditions associated with an extreme weather event

This paper extends the assessment of grid conditions under the extreme weather event into prevention state, degradation state, restoration state, and adaptation state. Fig. 3 shows the grid conditions associated with an extreme weather event. The vertical axis and horizontal axes represent the grid functionality the time, respectively. The grid is considered to be under normal conditions, Q_N, before an extreme weather event occurs at time t_E. An extreme weather event occurs at time t_E and causes the grid functionality to degrade, until it stops at time t_D, when the grid is considered at its worst condition $Q(t_D)$. This state will remain constant if no restoring actions are carried out, shown in Fig. 4 as degraded state. The grid functionality will gradually increase depending on the implementation of the restoring actions. The

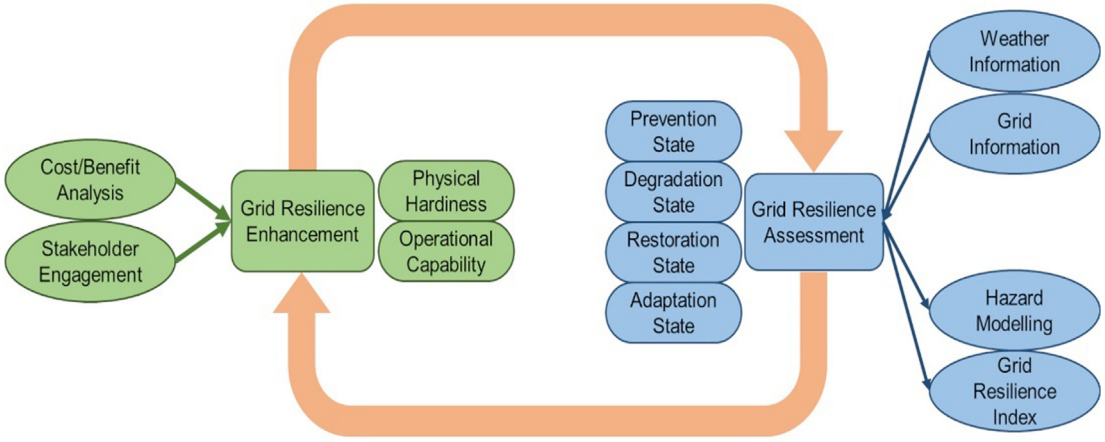


Fig. 2. The framework of grid resilience.

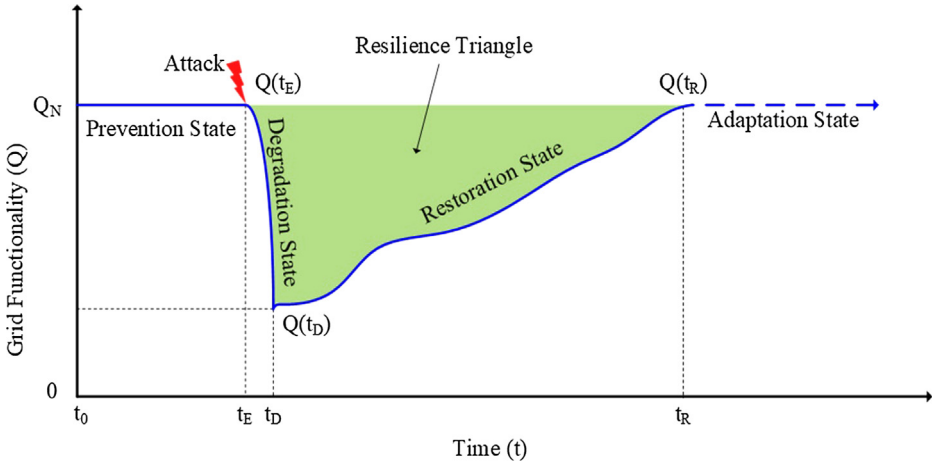


Fig. 3. Grid conditions associated with an extreme weather event.

restoration will continue until the grid achieves its pre-event condition at t_R ($Q(t_D) = Q_N$). At last, a continuous improvement must be achieved to face the next upcoming event.

4.1.1. State 1: prevention state

The grid is considered under normal operation and capable of providing its full functionalities in the prevention state. It is related to the short-term preparation to mitigate the impact caused by an upcoming extreme weather event. Therefore, the monitoring of the grid conditions is the key to improving grid resilience in this state [35]. It includes the estimation of possible conditions of the grids in various scenarios. The estimation of the grid can be performed using historical data to predict the damages, such as the number of component failures, the number of affected customers, and the duration of the damage [93]. By using the outcome of this estimation, some mitigation strategies can be deployed to prevent or to minimize the impact caused by the extreme weather event. For instance, if high precipitations are forecasted for the near future, actions such as installing a water barrier or flood walls [94], installing a waterproof enclosure for the grid components, or preparing a mobile and portable emergency generator or transformer [95,96], can be taken.

4.1.2. State 2: degradation state

The degradation state is the transition condition of the grid functionality from its normal to the damaged condition. The degradation state starts when an extreme weather event occurs, at time t_E, and ends when the grid reaches its worst condition at time t_D, as shown in Fig. 3. This state represents the magnitude of the impact caused by the occurrence of the extreme weather event. In addition, it also represents the physical strength of the grid, which contributes to the total functionality loss. If restoring actions are not taken immediately after or during the event, the grid functionality will remain in the degraded state.

The magnitude of the impact in the degradation state can be calculated by evaluating the failure state of the grid components during the event. The failure state of the grid components is determined by the physical failure or cascading failure. The physical failure occurs when the intensity of the extreme weather event exceeds the withstanding capability of the grid component. The damaged part may lead to cascading failure because the other parts of the grid will burden additional loads, which may be larger than their maximum loading capacity [97–103].

4.1.3. State 3: restoration state

The restoration state is the transition state of the grid from its damaged condition to its pre-event condition. It starts when a restoring action is deployed, at time t_D, and ends when the grid functionality achieves its pre-event state, as shown in Fig. 3. Unless this condition is attained, the grid is still considered in its restoration stage. This state represents the duration of the impacts caused by an extreme weather event and the dynamic characteristic of the grid.

The assessment is conducted by evaluating the restored state of the grid components since the restoration actions had started. A restoration action may bring back 100% of grid functionality or only a certain

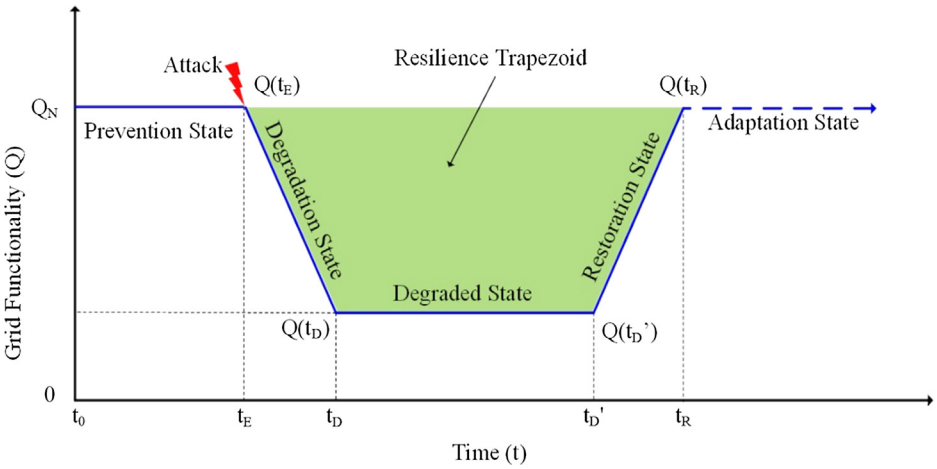


Fig. 4. Grid conditions associated with an extreme weather event in degraded state.

portion of it. For instance, damage on a substation may cause customers linked to that substation to lose the power, and utilization of a mobile substation may not be able to service all customer. Therefore, the operational capability of the grid is defined by two features, namely the restoration factor and the restoration time. The restoration factor is the portion (in percentage) of the grid functionality that can be restored by an operational action, and the restoration time is the time required for that operational action. These two features depend on the restoration resources of the grid, including the repair crew, spare parts, cost, access, transportation, and temporary power supply. Hence, the purpose of grid resilience enhancement at the restoration stage is on the improvement of its restoration time and operational agility.

4.1.4. State 4: adaptation state

The adaptation state is the fully restored state and occurs after the extreme weather event up to the next extreme weather event. The necessity of the analysis of this state along with the restoration state differentiates the study of grid resilience to the grid reliability. It starts immediately after the grid achieves its pre-event condition after being in the damaged condition, and ends when the upcoming extreme weather event is approaching. It can be considered as a long-term prevention state, which can be used to evaluate the latest event, identify the weaknesses of the grid, and implement the long-term enhancement strategies. This state must be considered in the study of grid resilience because it includes the ability of the grid to cope with similar events or even more extreme weather events in the future.

4.2. Quantification of grid resilience

Quantitative indices that can be applied to any power system grids are required to study grid resilience. The indices are used to evaluate grid performance, to compare between two or more grids, and to prepare and manage appropriate enhancement strategies. Good metrics have to meet specific criteria such as being quantifiable, repeatable, and comparable [104]. Although the study of resilience index in power systems is still limited, approaches used in various disciplines can be applied as references.

The quantification of grid reliability has been standardized which based on the frequency and the duration [31]. Whereas, quantification of grid resilience requires to delineate all the capabilities of a resilient grid such as anticipation, absorption, restoration, and adaptation. Furthermore, there is still no standard and single index that can represent all of the capabilities of a resilient grid. The straightforward way to define grid resilience is through investigation of the total impact. This is shown by the area between the start of the degradation stage up to the end of the restoration stage, which is called grid resilience triangle, as shown in green color in Fig. 3. Yodo and Wang [105] proposed the utilization of resilience triangle to measure the resilience indices. They extended the resilience triangle to calculate the severity of the impact, the profile length, and the weighted-sum, in addition to the performance loss and recovery period. Furthermore, the quantification of grid resilience was divided based on the total impact, pre- and post-event, and the restoration.

4.2.1. Grid resilience triangle and grid resilience trapezoid

The concept of resilience triangle was introduced by Bruneau et al. [25]. Since then, it became the foundation for the analytical assessment of resilience. Furthermore, this concept was extended to resilience trapezoid to consider the degraded state when no immediate restoration actions are taken upon disturbances [35,106]. Tierney and Bruneau [107] suggested that resilience can be measured through the state of functionality of a system after a disaster, and the time it takes to go back to the normal condition. They explained that the performance of any system is analyzed as one point in a multidimensional space of performance measures. This performance can be varied over time, either intentionally or unintentionally, suddenly or slowly and it can either increase or decrease. The unexpected and sudden decrease of system performance is regarded as the primary concern of resilience. Their work explained the basic concept of resilience assessment, which can be applied in any area of interest including power grid. In addition, Bocchini et al. [108] studied the resilience triangle in the civil infrastructure area. They interpreted as that the three legs of the triangle having a specific essence. The first leg (vertical) quantifies the functionality, the second leg (horizontal) expresses the total recovery time or rapidity, and the third leg (hypotenuse) defines the approximation of the functionality recovery path. On the other hand, D'Lima and Medda [109], who studied resilience in the transportation sector, stated that system resilience depends on the recovery speed, which is represented by the horizontal axis in the resilience triangle. They employed the mean-reversion model proposed by Uhlenbeck and Ornstein [110] to examine the rate of recovery.

According to Bruneau et al. [25], the impact can be measured by calculating the grid functionality difference between normal and abnormal condition during the occurrence of the event. It is simply the area of the resilience triangle, and mathematically is expressed by the following equation:

$$I = \int_{t_E}^{t_R} [Q_N - Q(t)]dt \tag{1}$$

where I is the impact caused by an extreme weather event, t_E and t_R are, respectively, the starting time of the event and at the time when restoration is completed, $Q(t)$ is the grid functionality at time t, and Q_N is the total grid functionality in normal condition. The grid functionality can be determined by the following equation:

$$Q(t) = \sum_{n}^{N} (\rho_n \times q_n) \tag{2}$$

where q_n is the service responsibility of the node n, which can be measured by its loading capacity. While ρ_n is the status of the node at time t, either normally operated or failed. This status is determined by the physical failure probability of the grid components, its maximum loading capacity, the restoration time, and the restoration factor. The physical failure probability of the grid components depends on their design and has mostly a lognormal distribution [111,112]. It can be expressed by the following equation:

$$P(t) = \Phi[\ln(S(t)/\theta)/\omega] \tag{3}$$

where $P(t)$ is the physical failure probability of a component, $S(t)$ is the weather intensity over the time of interest, θ and ω are, respectively, the median and the standard deviation of natural logarithm of the engineering parameter at which the component reaches the damage state thresholds, and Φ is the cumulative distribution function (CDF) of standard normal distribution. In a real condition, if the failure probability of a grid component is relatively high, it can be considered as failed. One approach to decide whether a grid component will fail or not is to make the pessimistic assumption that the feeder will fail if the physical failure probability is not zero. Another approach is to include the randomness of the event by comparing the physical failure probability with a uniformly random number between 0 and 1. If the failure probability is larger than the random number, it will be considered as failed, and vice versa [113].

The nodes and lines between the nodes have the maximum loading capacity. The physical failure of nodes or lines may affect the power delivery to the adjacent nodes via neighboring lines. When this power causes the loading of a node or line to exceed its capacity, they will be tripped. This process will continue until the power flow in the grid is balanced and stable. This phenomenon is known as cascading failure [113–115]. The loading of the nodes and lines can be retrieved through power flow analysis. Hence, any node or line that exceeds its maximum loading capacity is removed from the power flow calculation. After that, the power flow analysis is updated until no node or line is

overloaded. The final status of each node is used to calculate the grid functionality in Eq. (2).

The restoration time is the time required for a restoration action to bring back the grid functionality to its normal condition, while the restoration factor is the percentage of the service that can be regained by a restoration action. Full recovery of a grid component needs a longer time than partial recovery. For instance, the service obligation of a collapsed substation can be replaced partially and rapidly by a mobile substation, which, however, may only be able to cover some portion of the loads under the obligations of the collapsed substation. Restoration of the full obligation can be fully restored by rebuilding the collapsed substation, although it would require longer time. These two parameters determine the restored status of the nodes at a given moment during the event.

4.2.2. Grid resilience indices

Several methodologies to quantify the resilience index have been presented by some authors. In addition to the impact calculated in (1), some quantitative indices can be added to measure the grid resilience. Panteli et al. [106] introduced the $\phi\Lambda E\Pi$ (pronounced as "*FLEP*") resilience metrics to indicate the speed (ϕ) and the magnitude (Λ) of the damaged grid functionality, the duration of the damaged state (E), and the recovery speed (Π). Francis and Bekera [80] presented framework to calculate the resilience metric considering the recovery time and the system states. The damaged and restored states of the system are normalized to its stable state. It is expressed by the following equation:

$$R_C = \frac{Q(t_D)}{Q(t_E)} \times \frac{Q(t_R)}{Q(t_E)} \times Sp \tag{4}$$

where R_C is the grid capacity resilience index. This index considers absorptive, adaptive, and restorative capacity of the grid. $Q(t_R)$, $Q(t_D)$, and $Q(t_E)$ are the grid functionality at restored, damaged, and normal condition, respectively. Sp is the recovery speed index and is expressed by the following equation:

$$Sp = (t_D - t_E)/(t_R - t_E) \quad , \quad t_R \geq t_D \tag{5}$$

where t_E, t_R, and t_D are the times at the event start, at completed restoration, and at the worst damaged condition, respectively.

Another index was proposed by Henry and Ramirez-Marquez [91]. They compared the improvement of grid functionality from the damaged state with the reduction of functionality from the normal state. The full resilience is achieved when the restored grid functionality is equal to the normal (pre-event) grid functionality. It is expressed by the following equation:

$$R_Q(t) = \frac{Q(t) - Q(t_D)}{Q(t_E) - Q(t_D)} \tag{6}$$

where $R_Q(t)$ is the resilience index for grid functionality. The speed of restoration is not considered in this index.

Ouyang et al. [90] calculated the resilience index as a ratio between the real performance and the target performance of a system. Hence, the value of resilience is ranges from 0 to 1. The parameters used by this metric are the amount of flow or services delivered, the availability of critical facilities, the number of customers served, or the support of economic activities. They can be mathematically expressed in the grid resilience as in the following equation:

$$R_{inst} = \frac{\int_0^T Q_{actual}(t)dt}{\int_0^T Q_{target}(t)dt} \tag{7}$$

where R_{inst} is the instantaneous grid resilience index. $Q_{actual}(t)$ is the actual grid functionality, $Q_{target}(t)$ is the target grid functionality, and T is the observation period.

Shinozuka et al. [116] presented grid resilience in two separated indices. These indices represent the grid physical hardiness and operational capability. The indices are expressed by the following equations:

$$R_{physical} = Q(t_D) \tag{8}$$

$$R_{operational} = \frac{Q(t_R) - Q(t_D)}{t_R - t_D} \tag{9}$$

where $R_{physical}$ and $R_{operational}$ define the grid resilience index of the physical hardiness and operational capability during degradation stage and restoration stage, respectively. t_R and t_D are the time at the completed restoration and at the worst damaged condition. The index for grid physical hardiness is simply the amount of functionality that a grid can maintain during the event. The index for operational capability includes the amount of service restored from the damaged condition and the time required for that restoration.

Although there are difficulties in quantifying the grid functionality in the adaptation stage, it can be evaluated by analyzing the improvement on the grid parameters used in degradation and restoration stage. These parameters are physical hardiness (θ and ω), feeder maximum capacity, restoration factor, and restoration time. The grid is improved if these parameters are improved. Mathematically, they can be expressed in the following equation:

$$A = \frac{\Psi_{enhanced}}{\Psi_{pre-event}} \tag{10}$$

where A is the adaptability index, and Ψ represent the parameters used, such as physical hardiness, feeder maximum capacity, or restoration time. For instance, the adaptability improves if designing the components of the feeders is better than the one prior to the event, or the restoration time is faster than the restoration time prior to the event.

4.2.3. An example of grid resilience evaluation

The actual data during the period of two typical typhoon occurred at the same area in South Korea, such as typhoon Bolaven and typhoon Sanba, are used as an example to calculate the grid resilience indices. Typhoon Bolaven happened at the end of August 2012 and typhoon Sanba happened at the mid of September 2012 which is around three weeks after typhoon Bolaven. Typhoon Bolaven has average contained-wind speed of 126 km/h, the maximum radius is 450 km, the strength level is 4, the moving speed is 37 km/h, and the typhoon duration is 60 h. Whereas, typhoon Sanba has average contained-wind speed of 121 km/h, the maximum radius is 400 km, the strength level is 4, the moving speed is 39 km/h, and thee duration is 36 h.

The functionality of the grid was recorded during the occurrence of both typhoons based on the number of affected customers. The grid functionality differences between two typical event can be used to calculate the grid resilience index and to observe the improvement between through the analysis of the impact and the grid capability [83]. The grid functionality during typhoon Bolaven and typhoon Sanba is presented in Figs. 5 and 6, and the result of grid resilience index calculation is presented in Table 3.

It can be seen that the grid capacity resilience index (R_C) of a respected grid during typhoon Sanba is higher (0.45) compared with the grid resilience during typhoon Bolaven (0.09). The overall performance of the grid between two events is indicated by instantaneous grid resilience index (R_{inst}) of 1.61 which means that the grid resilience during the typhoon Sanba event has improved from the typhoon Bolaven event. Similarly, the physical resilience of the grid ($R_{physical}$) is 0.9 when typhoon Sanba occurred which is also higher than the physical resilience during typhoon Bolaven (0.35). Hence, the adaptability of the physical grid ($A_{physical}$) has improved by 1.61 times. However, the grid resilience of the operational capability ($R_{operational}$) during the latest event (typhoon Sanba) is lower than the previous event (typhoon Bolaven), such as indicated by the index of 0.03 compared with 0.07. Therefore, the adaptability of the grid operational ($A_{operational}$) is also decreased by 0.46 times.

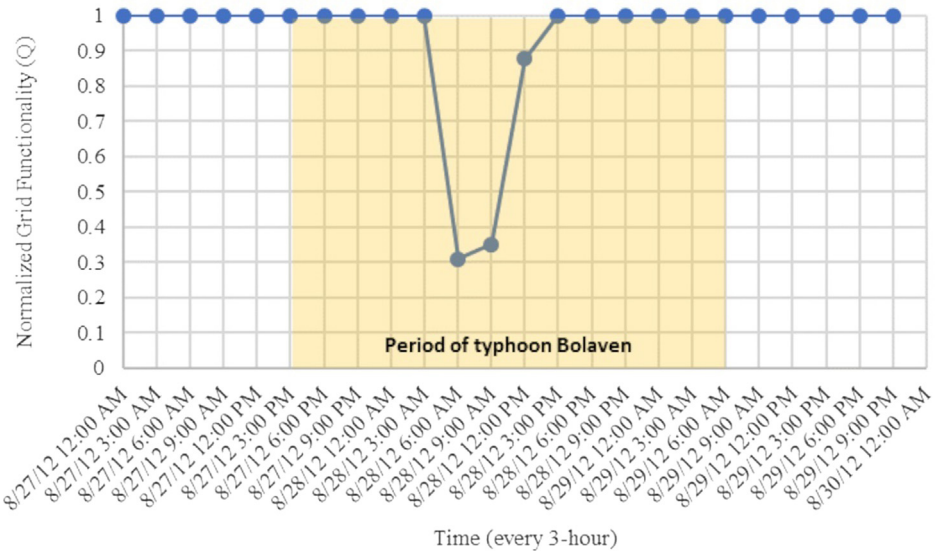


Fig. 5. Grid functionality during typhoon Bolaven in South Korea.

5. Grid resilience enhancement strategies

Grid resilience can be improved through the implementation of various enhancement strategies, which is in some of the literature, it is called grid hardening or grid adaptability. A careful decision to implement one enhancement strategy is required due to the limitations of the budget, resources, time, etc. Therefore, a specific characteristic of one utility grid must be prioritized for the implementation of grid resilience strategies.

Grid resilience enhancement strategies serve the two purposes: decreasing the magnitude of the immediate impact caused by an extreme weather event and restoring the grid functionality to its normal state as rapidly as possible after the extreme weather event. It can be categorized into two main groups such as the enhancement of grid physical hardiness and grid operational capability. Generally, the enhancement of grid physical hardiness is used to decrease the magnitude of the impact, and the enhancement of grid operational capability is applied to decrease the restoration time [117]. Table 4 provides a summary of previous studies on grid resilience enhancement strategies.

Table 3
Grid resilience indices of one area in South Korea.

Grid resilience index	Equation	Value	
		Typhoon Bolaven	Typhoon Sanba
R_C	$R_C = \frac{Q(t_D)}{Q(t_E)} \times \frac{Q(t_R)}{Q(t_E)} \times Sp$ $Sp = (t_D - t_E)/(t_R - t_E) \quad , t_R \geq t_D$	0.09	0.45
R_{inst}	$R_{inst} = \frac{\int_0^T Q_{actual}(t)dt}{\int_0^T Q_{target}(t)dt}$	1.61	
$R_{physical}$	$R_{physical} = Q(t_D)$	0.35	0.90
$A_{physical}$	$A = \frac{\Psi_{enhanced}}{\Psi_{pre-event}}$	2.57	
$R_{operational}$	$R_{operational} = \frac{Q(t_R) - Q(t_D)}{t_R - t_D}$	0.07	0.03
$A_{operational}$	$A = \frac{\Psi_{enhanced}}{\Psi_{pre-event}}$	0.46	

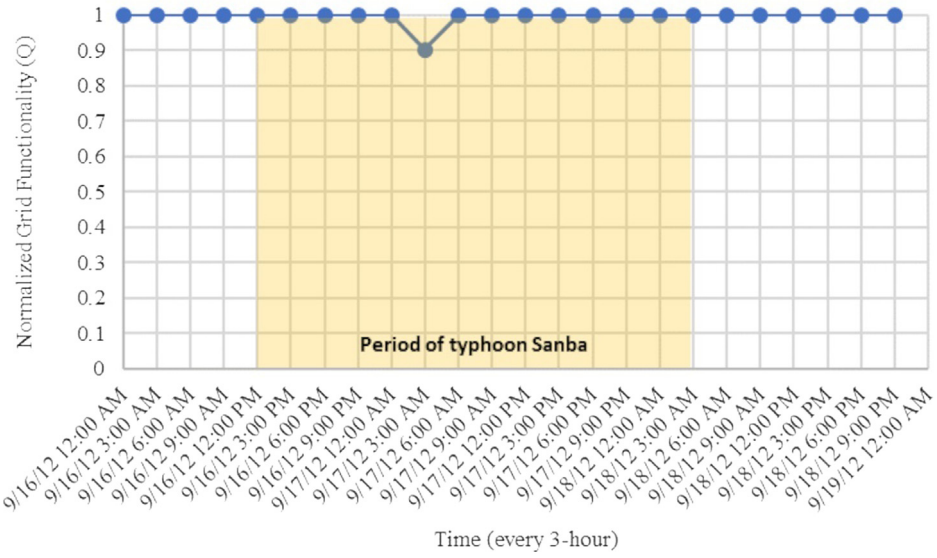


Fig. 6. Grid functionality during typhoon Sanba in South Korea.

Table 4
Classification of grid resilience enhancement strategies.

Grid capability	Enhancement strategy	Refs.
Physical hardiness	Vegetation management	[27,95,118–131]
	Selective undergrounding	[132–136,126,128,137]
	Physical upgrade and revitalization	[95,111,121,138–144]
	Elevated substation and water barrier	[49,82,95,122,130,145–147]
	Substation relocation and lines rerouting	[94,121,148–150]
Operational capability	Emergency generator and mobile substation	[1,27,95,96,121,130,151,152]
	Microgrids and DER	[7,124,146,153–165]
	Spare parts and repair crews management	[166–173]
	Grid monitoring system	[93,159,174–176]
	Regulation and design standards upgrade	[121,124,177,178]

5.1. Enhancement of grid physical hardiness

Grid physical hardiness is related to the static property of the grid. It acts as the first barrier to absorb the effect of the extreme weather events. The grid must be designed to withstand at least weather events of intensity equal to the intensity of the worst recorded weather event. The design purpose is to minimize the physical failure probability of the grid components.

5.1.1. Vegetation management

Vegetation management includes considering plans to plant new trees and pruning/trimming existing trees around the transmission and distribution lines. By doing so, it can minimize the airborne debris that touch the lines and cause short circuits and avoids fallen trees damaging the transmission, distribution lines, and poles during a storm or strong wind event. Vegetation planning includes the consideration of the appropriate distance between the lines and the trees, as well as the consideration of the access for repair equipment. The distance between the lines and the trees depends on the voltage the lines carry. For instance, high voltage transmission lines require more space than low voltage lines. Furthermore, the height of the trees closer to the lines must not exceed 10 feet [118].

Tree-trimming is considered the most widely implemented strategy in vegetation management. Conventionally, trees are trimmed considering the distance from trees and branches to the electricity lines. However, in the case of extreme weather events, debris can come from other areas due to strong winds. Therefore, other practices that can be applied to enhance the grid resilience during extreme weather events have been implemented, including the use of Geographic Information System (GIS), to manage the trees in a larger geographical area, sonic scanning, which can be used to get information from the most susceptible trees during the storm, and LIDAR (Light Imaging, Detection, and Ranging), to measure the conditions of the trees [119].

A scheduling algorithm for optimal vegetation maintenance for overhead line distribution systems was presented by Kuntz et al. [120]. They developed a model of vegetation-based failure rate which quantifies the reliability of distribution systems. The development of the scheduling algorithm had three objectives, namely minimizing the cost of reliability, minimizing the cost for a given reliability, and maximizing the reliability for a given cost subject to the vegetation maintenance crew availability, costs, and failure rate. A comprehensive summary of the practical implementation of vegetation management aimed at improving the grid resilience was provided by the Edison Electric Institute [121].

5.1.2. Selective undergrounding

Undergrounding the transmission and distribution lines serves to reduce the exposure of the poles and lines from the extreme weather events. This decreases the magnitude of the impact caused by an extreme weather event. However, a detailed study is required to decide whether this kind of hardening strategy is appropriate in the respective grid because the costs required to convert the overhead lines into underground lines is considerably high [132,135,136]. Studies found that undergrounding distribution system in a residential area in Florida, which includes the distribution network and main feeder, will cost an average of $13,500 per residential customer. The undergrounding cost for commercial areas on the other hand, will cost an average of $39,500 per commercial customer.

Another useful consideration when deciding to implement the undergrounding is the trade-off between the magnitude of the impact and the restoration time. Some studies showed that although undergrounding the power lines can provide benefits in the form of damage reduction, it may increase the restoration time because of the difficulties to trace the damaged lines. Therefore, a implementing a selective undergrounding strategy is more reasonable than complete undergrounding. Recent reports showed that selective undergrounding of the power lines, as opposed converting all overhead lines into underground lines, provided benefits for both damage reduction and costs [126,128,132–136].

A study by Kopsidas and Liu [137] presented a framework for evaluating the cable design and aging, which can be used in the selective undergrounding planning. The information provided by this framework is essential in the implementation of selective grounding because, if it is found that some underground lines are vulnerable to the upcoming event, an early preventive action can be taken before the extreme event occurs.

5.1.3. Physical upgrading and revitalization

The grid resilience can be improved through revitalization of the grid components. This strategy can be implemented by upgrading an old component's structure with the recent technology. Examples of upgrading are changing the power poles with stronger materials, applying a new pole design to withstand a stronger wind speed, and providing additional supports, such as guy wires and arm guy [95,121,138,139]. Ma et al. [111] studied grid enhancement by upgrading distribution poles and vegetation management and showed that a combination of both strategies was more effective compared with individual implementation. One algorithm for the improvement of the a model of the power poles physical strength was described by Shafieezadeh et al. [138] and Salman et al. [139]. The fragility of the power poles was assessed, and the necessary improvement or upgrade required by the poles was proposed based on the result of this assessment.

The earthquake-resistant building design can be applied to improve the physical hardiness of the substation, as explained in [140–142]. Utilization of the earthquake-resistant building will decrease the failure probability of the grid components. A model to determine the necessary actions required to improve grid resilience against earthquakes was presented by Navarro-Espinosa et al. [143].

5.1.4. Elevated substation and water barrier

The substation is vulnerable to floods caused by extreme weather events. One of the most practicable ways to maintain the substation

operation is to elevate it above the limit of the flood inundation level, and by installing water barriers around the perimeter of the substation [82,95,130,145,146]. Ideally, when floods are frequent in the substation, a fixed-type water barrier may be necessary. If the water level is forecasted to rise in the very near future, temporary water barriers may be installed.

5.1.5. Substation relocation and lines rerouting

Substation relocation and transmission and distribution lines rerouting may be an option to enhance the grid resilience despite its high costs of implementation [121,148]. If many critical consumers are supplied by a substation where, based on historical data, the occurrence of extreme weather is likely, relocating it and rerouting its lines becomes a priority [94,149]. A 2011 case study by Entergy Corp. on the power blackout caused by Hurricane Sandy and Irene, reassured that the relocation of the grid's components from the coastal area, is much more cost-effective than the implementation of other hardening strategies [150]. Therefore, a long-term cost-benefit analysis is required to decide the convenience of the substation relocation or lines rerouting.

5.2. Enhancement of grid operational capability

Grid operational capability is related to the dynamic properties of the grid, which means that the ability of the grid to manage the customer supply availability in a damaged grid condition is as high as possible. The fundamental principle of grid operational capability is to recover the grid functionality rapidly. There are two efficient ways to achieve this purpose. The first involves providing the temporary backup during the outage of the main power through an emergency power supply or reconfiguration of the grid's topology. The second involves conducting the repair and restoration of the damaged parts of the grid. Hence, the effectiveness of resource mobilization and the sequence of the restoration become the key of rapid recovery [179]. The resources include the backup power supply, spare equipment and spare parts, and repair crews. In addition, the enhancement of grid operational capability depends on the implementation of the operation and control algorithm for network configurations and automated self-healing algorithm.

5.2.1. Emergency generator and mobile substation

A conventional way to restore power supplies in a damaged area is to provide portable and mobile power generators or substations, as reported by [1,27,95,96,121,130,152]. The generation can be a diesel-driven or even a battery-type power generator. A mobile substation may consist of the power transformer, switchgear, and temporary control panel. The use of this equipment can maintain the electricity supply during emergency situations caused by extreme weather events, especially for remote but critical customers. It enables the temporary restoration of the grid functionality, in the case of the damaged substation or collapsed distribution lines, while a full restoration is under way. Furthermore, the emergency power supply can be used to support the black-start condition [151]. Another benefit of preparing the emergency generator and mobile substation is that it can be deployed earlier to the nearest and safest location where an extreme weather event is forecasted.

5.2.2. Microgrids and distributed energy resources (DER)

The development of the smart grids and microgrids have brought huge benefits to the grid in relation to extreme weather events, and their use have been expected to improve grid resilience [7,146,153,154,156,157,180]. They can provide an option to self-supply the local area when the main grid or branch feeders are damaged. The generation, storing, and control of the energy locally, can make the network less vulnerable to extreme weather events.

With smart grids and microgrids response to the effects of the extreme events can be faster and more efficient. Microgrids with defensive islanding can even provide an option to maximize the grid restoration [158]. This strategy enables the maximization of grid capacity by distributing the power to the customers. Moreover, the impacts of power transmission/distribution lines failure can be minimized by applying DER and microgrids. The application of the plug-in electric vehicles (PEV) and demand response (DR) through smart grids can also improve the grid resilience in relation to extreme weather events [159].

Campbell [124] has made various efforts to reduce storm-related power outages, including the implementation of smart grids and distributed generation (DG). Smart grids will help resolving power outages issues because they can detect problems and manage alternatives by reconfiguring the topology while alerting the operators about the outages and their locations at the same time. Distributed generation is located close to the loads so that its power is delivered through the minimum number of transmission or distribution lines. Therefore, the use of the DG will reduce the power outages at the customer site due to the failure of transmission/distribution lines caused by extreme weather events.

Although the implementation of the microgrids and DER may improve the grid resilience, more studies and tests are required to determine their optimal size, location, and operation. Qiu et al. [160] presented a framework for power system network expansion, which can be used to determine a more resilient network configuration in the expansion planning. The optimal solution was obtained by solving the objective function to minimize the total cost and the risk level of unacceptable reliability. The total cost included the investment cost and operational cost. The risk level of unacceptable reliability was determined based on the load curtailment. Zhang et al. [161] presented an optimal placement, as well as an optimal sizing, of DG such as PV generation and battery storage to enhance grid resilience. However, the objective of the strategy was to reduce the Energy Not Supply (ENS) by considering the investment and operational cost of DG, the access to the demand, and the access to the non-black-start generation units, which usually provide a larger amount of generation capacity. A similar approach was taken by Zare et al. [162] to locate the optimal switches to improve the grid resilience against hurricanes. By implementing the fragility curve and *N-K* failure scenario, the algorithm used the model to obtain the impacts caused by the hurricane. The optimal placement of the switches was determined by minimizing the interrupted load of each configuration of the time step, normal feeder, damaged feeder, load points, and customer type. Dehghanian and Aslan [163] also considered the control of the grid topology to enhance the grid resilience by including the forecasted weather information and implementing the risk management strategy. This method can be used to prepare the mitigation plan when facing an upcoming extreme weather event by controlling the Transmission Line Switching (TLS) prior to the event.

Amirioun et al. [164] presented the scheduling strategy of microgrids to deal with the upcoming floods. The proposed strategy includes a vulnerability study, a worst-case analysis, and a proactive scheduling. The vulnerability study was used to identify the vulnerable components by applying the information of flood forecasting, flood characteristics, and the fragility of the grid components. The worst-case analysis was performed to identify the characteristics of microgrids under worst-case contingency by applying an optimal power flow. The proactive scheduling was implemented to find the configuration of microgrids that minimizes the load curtailment during the worst-case condition. Furthermore, this methodology can also be applied to other types of extreme weather events other than floods. Wang et al. [165] also simulated the proactive operation of the grid by considering a Markov process to minimize the current and future costs caused by the damages. First, the fragility curve of the grid components was modeled specifically for different types of extreme weather, and it was used to calculate the probability of grid components damage. The current and future states of the grid due to the damaged component were defined and used to determine the operation of the grid. Hence, the decision of grid operation was based on its current and future cost.

5.2.3. Spare part and repair crew management

Availability of spare parts is crucial to decrease the restoration time during the extreme weather event [166]. Due to the complexity of the grid, when preparing spare parts, many factors must be considered. There are many components that make up the utility, including the generation, transmission, and distribution systems. However, the decision can be made, and the priorities can be understood, by analyzing the component's failure rates, consequences, investment cost, and the operation and installment difficulty level [167].

As well as the availability of the spare parts, their location, their ease to access damaged area, and the number of repair crews must also be considered. Studies have recognized that inadequate spare parts and limited access to the location of the damaged area have caused longer restoration time and even led to larger damage [168,169]. Arif et al. [170] presented an optimization technique for repair crew routes to accelerate grid restoration. This technique included the optimal placement of the central station, which can be used to locate the spare parts and repair crews, and the determination of the dispatching route, that maximizes the pick-up loads. Hindolo et al. [181] proposed a maintenance strategy for complex power systems using a semi-Markov stochastic process. The maintenance process was divided into Preventive Maintenance (PM) and Corrective Maintenance (CM), which can be scheduled either at the same time or on their own and by considering the optimized sharing of maintenance crew in the nodes. Different methodologies for restoration scheduling by dispatching the repair crews were also discussed in [171–173].

5.2.4. Extreme-weather-event-based grid monitoring system

The actual and real-time information is important to quickly and properly plan and deploy the restoration actions during extreme weather events. However, due to the extreme weather event, the communication channel, which is used to share the information, may be out of order. Therefore, it is necessary to provide alternative solutions for receiving and broadcasting information. An example are unmanned aerial vehicles (UAV) to support Airborne Damage Assessment Module (ADAM), which will help to reduce response time and costs [159]. The ADAM will support the outage management system, geographical information system, and asset management system. The Electric Power Research Institute (EPRI) is developing a data visualization technology for utility engineering and field operations. It will allow the operators to retrieve real-time data about the utility poles and lines via computer, tablet, or smartphone by simply aiming the camera at the equipment to be inspected. The disaster area can be surveyed and the possible shortest route to deploy the repair and reinforcement team can be determined based on the information provided by this equipment. Another advantage of this method is that the drone can reach areas that cannot be reached by other vehicles, especially in cases such as when roads are blocked. Grid condition monitoring can also improve the situational awareness, which will lead to the improvement of grid resilience [174,175]. The effective responses by the system operator and other related parties, towing to their awareness of the latest conditions, can preserve the functionalities of the grid.

Monitoring the grid condition associated with an extreme weather event can also be achieved through damage estimation [93,176]. historical data of extreme weather events and damages of the grid are used to develop models to predict the impacts of upcoming events. A machine learning-based classification algorithm can be implemented to predict the number of failed components, failure duration, and the number of affected customers. The results of this algorithm can be used to plan necessary actions to minimize the damages, such as preparing the spare parts, increasing the number of repair crews, restoring schedule management, and switching sectionalizing management.

5.2.5. Regulation and design standards upgrade

The implementation of regulations and management during the disrupting events will determine the time required for recovery, as well as the cost, so proper regulations and management are necessary to improve grid resilience [121,124]. The mutual assistance agreement is a pre-event agreement between utility operators and other sectors. This agreement states that, during extreme weather events, the utility operator will be prioritized regarding support from other sectors. This will shorten the restoration time when the grid requires support from other parties.

The rise of the extreme weather events severity enforces the modification and upgrade of the current design and engineering standards, as well as the operation and maintenance procedures [124,177,178]. The analysis of the existing grid components and the severity of the upcoming extreme weather events must also lead to suggestions on how to improve the design in order to face more extreme weather in the future. Hence, the proposed design of the grid components can be implemented when the grid is expanded.

Another consideration when selecting the enhancement strategies is to ensure that implementation of one enhancement strategy will not downgrade other capabilities of the grid. For instance, undergrounding the cables may improve the physical hardiness, and thus minimize the initial impact during the storms, while increasing the recovery time if it is damaged during the floods. Therefore, the multi-events assessment must be considered to achieve the most effective and efficient solutions.

The implementation of grid resilience enhancement strategies will improve the grid capability but it requires large investments. On the other hand, a fragile grid may turn into huge economic loss due to failures in grid functionality. Therefore, optimal selection of the enhancement strategies is required based on cost/benefit analysis.

Apart from cost/benefit analysis, grid resilience enhancement strategies must consider the criticality of the grid parts (feeder). The importance of grid parts can be classified using Reliability Achievement Worth (RAW) and Reliability Reduction Worth (RRW) [17,182–185]. The RAW of one feeder is the ratio of resilience when that feeder is fully functioning to the actual resilience. The RRW of one feeder is the ratio of the actual resilience to the resilience when that feeder is failed. Based on this index, the feeder can be ranked from the most critically important to the less critically important.

6. Conclusion and future works

6.1. Conclusion

This paper presents a state-of-the-art review of the grid resilience studies that focuses on the technical matters from the point of view of power system engineering. It includes clarifications on grid resilience definitions, assessment and quantification, and enhancement strategies. Grid resilience is defined as the anticipative, absorptive, restorative, and adaptive capacity of the grid subjected to impacts caused by extreme weather events. Grid resilience is differentiated from grid reliability by the type of events and the behavior of the grid to those events. A grid resilience framework is used to provide a general scheme to understand the subjects in the study of grid resilience. The framework includes grid resilience assessment, quantification, and enhancement, thus covering all the main aspects of grid resilience.

Grid resilience can be assessed based on the impact or the grid capability. Depending on the grid conditions as a result of the extreme weather event, the assessment can be divided into prevention stage, degradation stage, restoration stage, and adaptation stage. Several indices are also presented to quantitatively measure grid resilience. However, it is worth noting that the quantification of grid resilience requires more than a single index, given the requirement to evaluate the magnitude of the grid functionality as well as the recovery time.

The purpose of the grid resilience enhancement strategies is to decrease the functionality losses and to shorten the restoration time. Various grid resilience enhancement strategies are presented based on grid physical hardiness and grid operational capability. Enhancement of grid physical hardiness serves to reduce the magnitude of the impacts,

while enhancement of the grid operational capability aims at reducing the restoration time. The categorization of the grid capability enables proper decision and implementation of the enhancement strategies.

6.2. Future works

Although this paper serves a comprehensive review on grid resilience, some limitations are not considered and need to be discussed in the future. Further studies required to address these limitations in the study of grid resilience are listed:

- The definition of resilience of the power system is taken from definitions provided by other disciplines. The standardization of grid resilience definition is required, as a basis for further study.
- The indices that were presented are only describing the absorptive and restorative capacity of the grid. Indices that describe the anticipative and adaptive capacity of the grid are also required.
- As described in Section 5, various enhancement strategies have been developed, but the modeling study for those strategies is still limited. The development of models and the related parameters for each strategy is required to assess the effects of those strategies on grid resilience.
- The comparison and optimization analysis for grid resilience enhancement strategies is necessary. The analysis must include cost/benefit considerations and stakeholder preferences. The results from this analysis are used to decide which strategy should be implemented on one grid.
- Multi-event assessment is required to ensure that the enhancement strategy related to one extreme weather event does not deteriorate the grid capability related to another extreme weather event.

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